MIL-P-63194A (AR)
19 November 1981
SUPERSEDING
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18 October 1977

MILITARY SPECIFICATION

PROPELLANT, M10 FOR 60MM MORTAR M204 PROPELLING CHARGE

This specification is approved for use by the U. S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers the requirements, quality assurance provisions, and packaging of M10 Propellant for use in M204 Mortar Increments for the 60mm cartridge: HE, M720.

1.2 Classification. The propellant shall be M10 Flake.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. Unless otherwise specified (see 6.1), the following specifications and standards of issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-414 - Sampling Procedures and Tables For Inspection By Variables For Percent Defective

MIL-STD-652 - Propellants, Solid, for Cannons, Requirements and Packing

2.1.2 Other Government documents, drawings, and publications. The following documents and drawings form a part of this specification to the extent specified herein.

DRAWINGS (see 6.9)

U. S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9312698 - Charge, Propelling, M204
9275526 - Cartridge, 60mm: HE, M720
9312697 - Container Assembly

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. The propellant shall be M10, conforming to the requirements of MIL-STD-652 and drawing 9312698 and with a nominal thickness of .009 inches.

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3.2 Ballistic assessment requirements. The propellant when loaded into the M204 container assembly and drop fired in the M224 mortar and with the weight of projectile (M720, 3.75 ± .01 (lbs)) for which it was manufactured, shall comply with the following requirements when tested at 70°F and standard weapon conditions (see 6.11). The propellant shall be capable of assessment to a velocity of 795 feet per second (fps) with an expected average pressure not to exceed 7580 pounds per square inch (psi).

3.3 Ballistic uniformity requirements. The propellant when fired in the M224 mortar at the calculated charge weight and with the projectile M720 (see 6.5) for which it was manufactured, shall comply with the following requirements when tested as specified in 4.5.1.2.

3.3.1 Uniformity of velocity. The standard deviation of velocity for the lot at 70°F shall not exceed 6.0 fps.

3.3.2 Average pressure At standard weapon conditions and recommended charge weight, the propellant shall not produce an average pressure greater than 7580 psi at 70°F and 8590 psi at 145°F.

3.3.3 Individual maximum pressure. The propellant shall not produce an individual pressure greater than 10,080 psi at 145°F when fired in a standard weapon (see 6.10).

3.4 Preproduction sample inspection. This specification contains provisions for preproduction sample inspection. Requirements for the submission of preproduction samples by the contractor shall be as specified in the contract.

3.5 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. Preproduction sample inspection and quality conformance inspection shall be conducted on this item.

4.3 Preproduction sample inspection

4.3.1 Submission. The contractor shall submit a preproduction sample as designated by the contracting officer for evaluation in accordance with 4.3.2. The preproduction samples shall consist of three (3) pilot lots, approximately 50 lbs each. A new sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity, loadability, or ballistic performance as determined by the Technical Agency (see 6.12).

4.3.2 Inspections to be performed. Each of the pilot lots comprising the preproduction sample and submitted in accordance with 4.3.1 may be subjected by the Government to any or all of the requirements of the applicable drawings and specifications.

4.3.3 Rejection. If none of the pilot lots submitted meets the requirements of this specification, a new preproduction sample shall be submitted in accordance with 4.3.1. A pilot lot which meets the requirements of the applicable drawings and specifications may be designated by the contracting officer as the accepted preproduction sample. Until the preproduction sample is accepted, the contractor is in no way authorized by the Government to initiate regular production, unless otherwise directed by the contracting officer. All future production lots should be of the same basic granulation and process as the accepted pilot lot.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

PRECAUTION - This specification covers sampling and testing of chemical toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.2 Examination and testing

4.4.2.1 Sampling for laboratory testing (see 3.1). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical/physical properties in accordance with MIL-STD-652.

4.4.2.2 Sampling for ballistic testing (see 3.2). Each lot of propellant shall be sampled for Proving Ground tests in accordance with MIL-STD-652.

4.4.2.3 Charge establishment (see 3.2). If the lot does not meet the applicable requirements, testing shall cease and the test director shall immediately notify the contracting office.

4.4.2.4 Velocity uniformity (70°F) (see 3.3.1). If the product of the factor (0.73) times the velocity standard deviaton of the rounds tested in accordance with 4.5.1.2 exceeds the applicable requirement, apply retest and referral criteria in accordance with 4.5.1.3 and 4.5.1.4. If the retest and referral criteria do not apply, the lot shall be rejected.

4.4.2.5 Average pressure (see 3.3.2). If any average pressure obtained during initial or subsequent production at 70°F or 145°F, when corrected to standard conditions and recommended charge weight, exceeds the applicable requirement, apply retest and referral criteria in accordance with 4.5.1.3 and 4.5.1.4. If the retest and referral criteria do not apply, the lot shall be rejected.

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4.4.2.6 Individual maximum pressure (initial production). If any individual pressure observed during the 145°F uniformity series, when corrected to standard conditions and recommended charge weight, exceeds 10,080 psi, the lot shall be rejected.

4.4.2.7 Individual maximum pressure (subsequent production). If any individual pressure observed during the 70°F uniformity series, when corrected to standard conditions and recommended charge weight, exceeds the 70°F individual maximum pressure (10,080 psi minus the quantity of 75 times the pressure temperature coefficient established during initial production testing), and calibration round performance is acceptable (see 4.5.1.4), the lot shall be tested at 145°F in accordance with 4.5.1.2.1. If any individual pressure at 145°F, when corrected to standard conditions and recommended charge weight, exceeds 10,080 psi, testing shall cease and the lot shall be rejected.

4.5 Test methods and procedures

4.5.1 Proving ground assessments. This test shall be performed at a Government Proving Ground in accordance with the applicable acceptance test procedure using M720 HE projectiles inert loaded within 0.01 lb of an as fired weight of 3.75 lb and fired from M224 Mortar tube on a hard mount at an elevation of 800 mils. All charges shall be temperature conditioned in a large conditioning box, with adequate air circulation to assure uniform conditioning, at the required temperature for a minimum of 24 hours and assurance will be made that, when fired, the charges are at the conditioning temperature conditionings: -50°F, +5°F; 70°F, ±2.5°F; 145°F, -5°F. Copper crusher gages (2) shall be used to measure the maximum chamber pressure of each charge. Velocity coils will be used to measure the muzzle velocity of each charge. The container assembly lot number of the test charges shall also be recorded.

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4.5.1.1 Charge establishment

4.5.1.1.1 Charge slope method. Three or more groups of rounds, three rounds for each group loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F. Three calibration rounds shall also be conditioned and fired at 70°F, prior to the test rounds. From the data obtained, a composite charge weight versus velocity versus pressure curve shall be established. A charge weight shall be calculated which will yield the prescribed service velocity (see 3.2). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.5.1.2). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously estabished. The result will be the recommended charge weight.

4.5.1.1.2 Constant slope method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite charge weight versus velocity versus pressure curve can be established from test results generated from previous ten lots. This curve can then be used as in 4.5.1.1.1 to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as above for the charge slope method. The resultant will be the recommended charge weight.

4.5.1.2 Uniformity

4.5.1.2.1 Initial production. Thirty rounds shall be assembled with the calculated charge weight of propellant. Ten rounds each shall be conditioned to -50°F, 70°F and 145°F (as indicated by thermocoupled control round). Rounds fired at 70°F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibration performance under test conditions and standard conditions). Firing results at all temperatures shall be corrected to standard projectile weight (See 3.2), and to recommended charge weight conditions (i.e., the difference between as fired uniformity charge weight and recommended charge weight).

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The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results. The pressure-temperature coefficient for the lot shall be calculated from the pressure data obtained at 70°F and 145°F. The average of the coefficient for ten lots shall be utilized to calculate at 70°F individual maximum pressure for use in the subsequent production.

4.5.1.2.2 Subsequent production. Ten (10) rounds shall be assembled and tested at 70°F, fired alternately with calibration rounds. The means and standard deviations of the velocity and pressure shall be calculated on results corrected to standard conditons and recommended charge weight. If any individual pressure exceeds the 70°F individual maximum pressure (see 4.4.2.7), ten additional test rounds shall be conditioned and fired at 145°F. All results at 145°F shall be corrected to standard projectile weight (see 3.2). Throughout production, if 3 consecutive lots fail on first test, testing shall revert back to the initial production plan (see 4.5.1.2.1).

4.5.1.3 Retest criteria. A lot shall be retested whenever any of the following conditions exist or whenever the proving ground deems it necessary.

a. The final adjustment in charge weight exceeds 2% of the charge weight used in uniformity firing, retest for charge establishment and uniformity at 70°F.

b. If for any reason, the proving ground considers that test conditions have detrimentally affected the test results, additional rounds as required shall be tested.

4.5.1.4 Referral criteria. A test lot shall be referred to the procuring activity for disposition whenever any of the following conditions exists or whenever the proving ground deems it necessary.

a. Calibration rounds fired in any test phase fail to comply with Section 3 of this specification.

b. Calibration correction applied exceeds 15.9 fps of the expected muzzle velocity.

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c. Calibration components are not available.

d. Number of test or calibration results utilized in calculations is less than specified.

e. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, stickers, or projectile flights, etc. Each such event shall be reported in detail on the applicable firing record.

f. Test lot fails velocity standard deviation or average pressure requirement at 70°F and any extreme test value is an outlier. The following test shall be performed to determine if a single test result is an outlier. With a sample size of 10 at a significance level of 0.05, the critical value (T_C) is 0.477. Calculate a test value (T_t) as follows:

For Low Outlier

$$T_t = \frac{(X_2 - X_1)}{(X_9 - X_1)}$$

For High Outlier

$$T_t = \frac{(X_{10} - X_9)}{(X_{10} - X_2)}$$

Where:

X_1 = test result being tested as low outlier
X_2 = test result adjacent to X_1, after all the test results are ordered from smallest to largest
X_9 = test result adjacent to the largest test result
X_{10} = test result being tested as high outlier

X_1 or X_{10} is an outlier if T_t is greater than T_C

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652, except that all propellant designated for use as reference shall be packed Level A.

6. NOTE

6.1 Ordering data. Ordering data shall be in accordance with MIL-A-48078 with the following exceptions.

6.1.1 Acquisition requirements. Acquisition requirements of MIL-A-48078 shall apply.

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6.1.2. Contract data requiremnts One copy each of the propellant Description sheet and the Acceptance Test Summary shall be forwarded to each office designated by the Contracting Officer. Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity.

6.2 Standard deviaton. The standard deviation shall be calculated with (n-1) as the divisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, Example B-1.

6.3. Initial production lots. The initial production lots represent the first ten consecutive lots from each manufacturer which have been produced and meet the acceptance criteria of this specification. Throughout production, if 3 consecutive lots fail on first test or a lapse in production of 90 days or more occurs, testing shall revert back to initial production plan (see 4.5.1.2.1).

6.4 Surveillance sample. Samples shall be forwarded to U. S. Army R&D Command, Dover, NJ 07801, ATTN: DRDAR-LCE-MP for the propellant surveillance test program in accordance with MIL-STD-652.

6.5 Test rounds. Test rounds shall be assembled in accordance with drawings 9312698, 9275526, and 9312697.

6.6. Proving ground test summary

Temp	Sample Size		Test		Results	
			Velocity (fps)		Pressure (psi)	
°F	Cal	Test	Avg	Std dev	Ind Max	Avg
			Charge Establishment			
70	3	9	795	-	-	7580
			Uniformity (initial Production)			
-50	-	10	-	-	-	-
70	10	10	795	6.0	-	7580
145	-	10	-	-	10,080	8590
			Uniformity (Subsequent Production)			
70	10	10	795	6.0	see 4.5.1.2.2	7580
145*	-	10	-		10,080	8590

*To be fired only when 70°F results indicate that marginal pressure performance may exist (see 4.5.1.2.2 and 4.4.2.7)

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6.7 Retest. All original and retest data shall be submitted to U.S. Army R&D Command, ATTN: DRDAR-QAR-R and DRDAR-LCA-G, Dover, NJ 07801.

6.8 Test anomalies. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, stickers, poor projectile flights, etc., shall be reported in detail on the applicable firing record.

6.9 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal. Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.10 Individual Maximum Pressure Basis The individual maximum pressure value for 145°F cited is based upon the weapon design pressure.

6.11 Standard weapon. Standard weapon and standard weapon conditions refer to the weapon model and conditons of the test at the time the master calibration series was fired.

6.12 Preproduction sample. The preproduction sample is intended to provide the contractor with the necessary latitude to establish the propellant granulation needed to meet the required ballistic parameters. The Technical Agency and the contractor should jointly select the one pilot lot which best represents the desired parameters. All future production lots should be of the same basic granulation as the accepted pilot lot.

6.13 Reference sample. Whenever a propellant lot is selected for reference use, a 5- lb sample shall be forwarded to U.S. Army Armament R&D Command, Dover, NJ 07801, ATTN: DRDAR-LCA-G for use in evaluating the propellant stockpile.

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